EXP NO: 4A

SUPPORT VECTOR MACHINES (SVM)

AIM

To build an SVM model for a binary classification task, tune its hyperparameters, and evaluate it using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC.

PROCEDURE

1. Import the required libraries: NumPy, pandas, Matplotlib, and scikit-learn.
2. Load the Breast Cancer Wisconsin dataset from sklearn.datasets for binary classification.
3. Split the dataset into training and testing sets using an 80/20 split with a fixed random_state.
4. Standardize the features using StandardScaler, because SVMs are sensitive to feature scale.
5. Define an SVC model with an RBF kernel.
6. Tune the C and gamma hyperparameters using GridSearchCV with five-fold cross-validation and F1-score.
7. Train the final model using the best parameters and generate predictions for the test set.
8. Evaluate the model using accuracy, precision, recall, F1-score, confusion matrix, classification report, and ROC-AUC.
9. Plot the ROC curve and report the best parameters, evaluation metrics, and observations.

CODE

```
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    confusion_matrix, classification_report, roc_auc_score, roc_curve
)
from sklearn.datasets import load_breast_cancer
import pandas as pd
import matplotlib.pyplot as plt
import warnings

warnings.filterwarnings("ignore", category=FutureWarning)

# 2) Load dataset (binary classification)
data = load_breast_cancer()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target, name="target")  # 0 = malignant, 1 = benign

# 3) Train/test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

# 4) Standardize features (important for SVMs)
scaler = StandardScaler()
X_train_sc = scaler.fit_transform(X_train)
X_test_sc = scaler.transform(X_test)

# 5) Define model
svm = SVC(kernel='rbf', probability=True, random_state=42)

# 6) Hyperparameter grid & tuning
param_grid = {
    "C": [0.1, 1, 10, 100],
```

```
    "gamma": ["scale", 0.01, 0.001, 0.0001],
}
grid = GridSearchCV(
    estimator=svm, param_grid=param_grid, scoring='f1',
    cv=5, n_jobs=1, verbose=0
)
grid.fit(X_train_sc, y_train)
print("Best Parameters from Grid Search:", grid.best_params_)
best_svm = grid.best_estimator_

# 7) Train final model & predict
best_svm.fit(X_train_sc, y_train)
y_pred = best_svm.predict(X_test_sc)
y_prob = best_svm.predict_proba(X_test_sc)[:, 1]

# 8) Evaluation
acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred, zero_division=0)
rec = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
auc = roc_auc_score(y_test, y_prob)
cm = confusion_matrix(y_test, y_pred)

print("\n=== SVM (RBF) – Test Metrics ===")
print(f"Accuracy : {acc:.4f}")
print(f"Precision: {prec:.4f}")
print(f"Recall : {rec:.4f}")
print(f"F1-Score : {f1:.4f}")
print(f"ROC-AUC : {auc:.4f}")

print("\nConfusion Matrix:\n", cm)
print("\nClassification Report:\n", classification_report(y_test, y_pred, zero_division=0))

# 9) Plot ROC Curve
fpr, tpr, thresholds = roc_curve(y_test, y_prob)
plt.figure()
plt.plot(fpr, tpr, label=f"SVM (AUC = {auc:.3f})")
plt.plot([0, 1], [0, 1], linestyle="--", color='gray')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve – SVM (RBF)")
plt.legend()
plt.grid(True)
plt.show()
```

OUTPUT

Actual execution output:

```
Best Parameters from Grid Search: {'C': 10, 'gamma': 0.01}

=== SVM (RBF) – Test Metrics ===
Accuracy : 0.9825
Precision: 0.9861
Recall : 0.9861
F1-Score : 0.9861
ROC-AUC : 0.9977

Confusion Matrix:
 [[41  1]
 [ 1 71]]

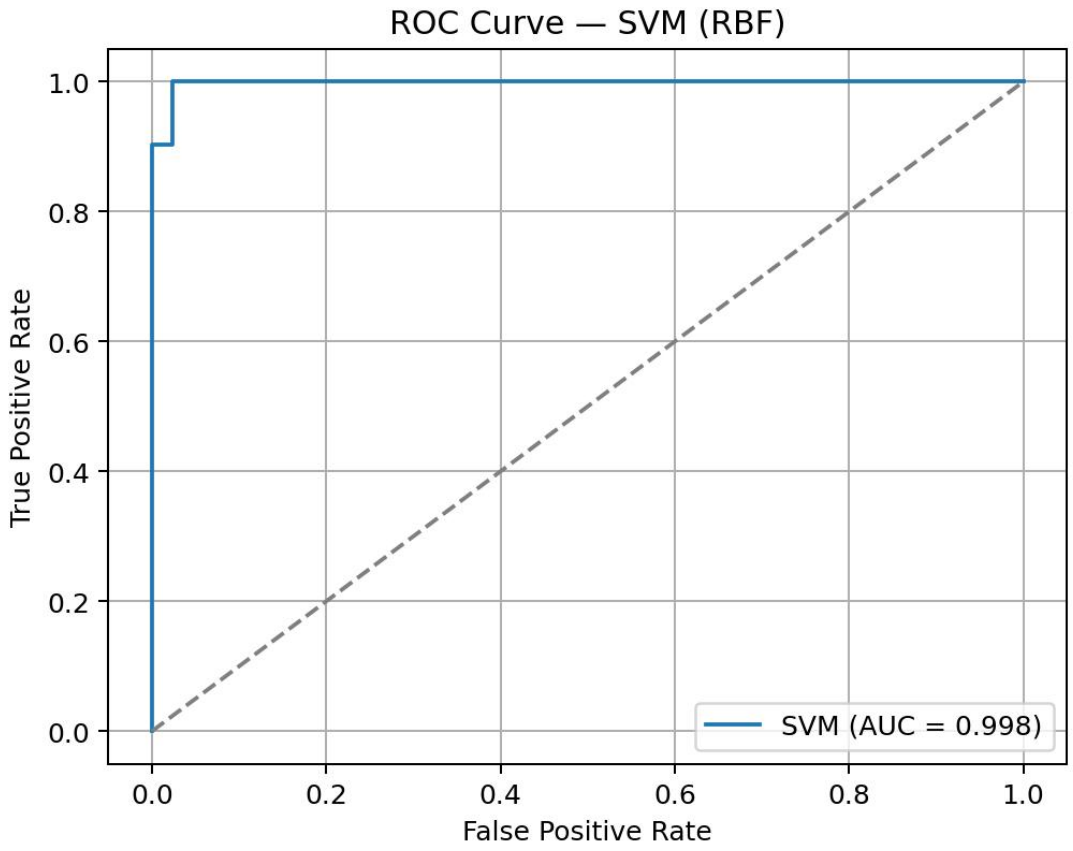
Classification Report:
              precision    recall  f1-score   support
```

```
              precision    recall  f1-score   support

           0       0.98      0.98      0.98        42
           1       0.99      0.99      0.99        72

    accuracy                           0.98       114
   macro avg       0.98      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114
```

ROC Curve — SVM (RBF)



RESULT

The Support Vector Machine (SVM) model was successfully implemented and evaluated on the Breast Cancer Wisconsin dataset. The model was tuned using GridSearchCV, and its performance was evaluated on the test set using accuracy, precision, recall, F1-score, the confusion matrix, the classification report, and ROC-AUC. The ROC curve provides a visual representation of the model's ability to distinguish between the two classes.